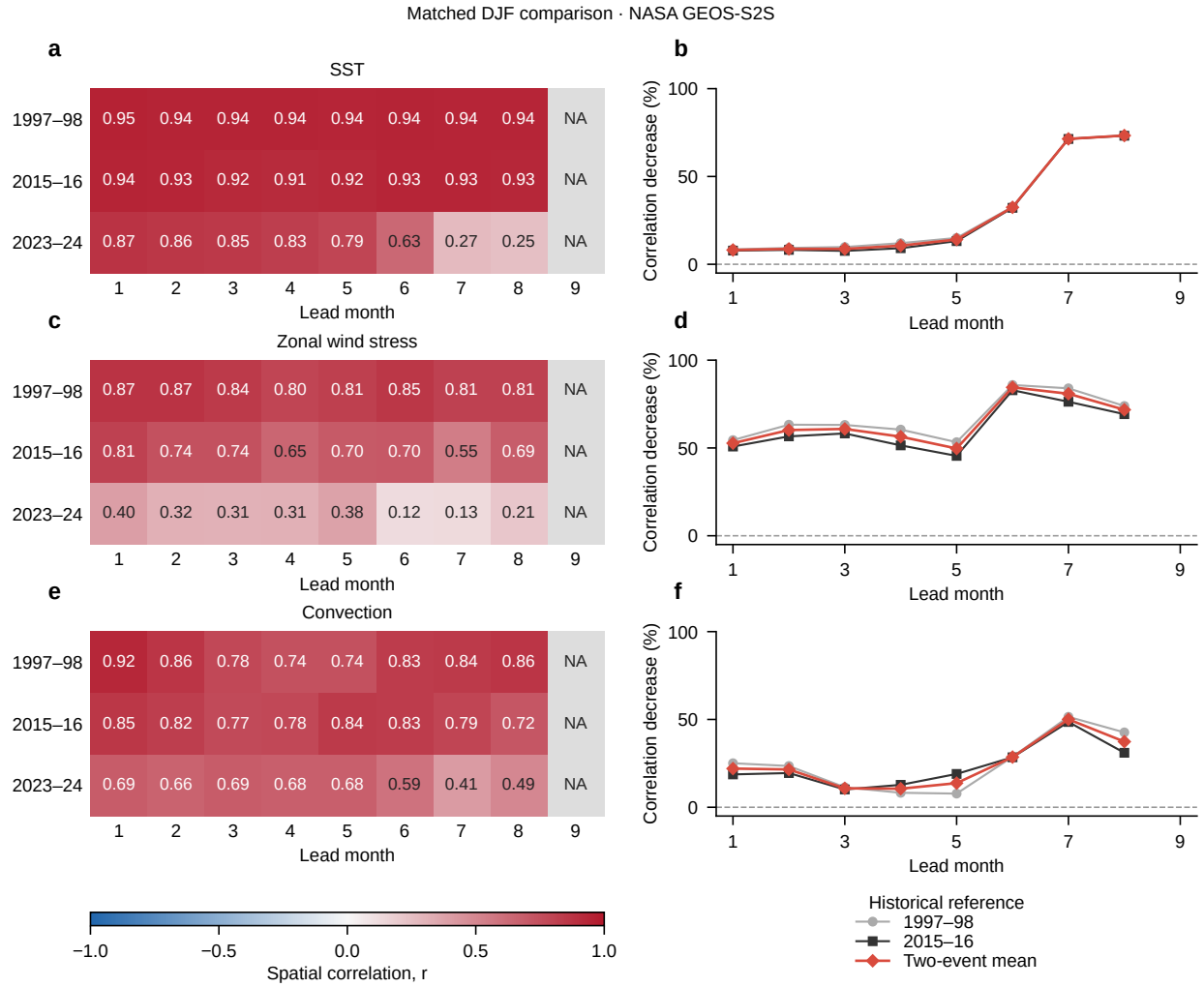
Supplementary Fig. 5 | Tropical forecast skill under matched model, season and lead conditions



a,c,e, Cosine-latitude-weighted spatial correlations between forecast and observed DJF anomalies for SST, zonal wind stress and convection, represented by precipitation. All variables use NASA GEOS-S2S, the same four ensemble members (1–4), the same target months and leads, and the same tropical Pacific ocean mask (20°S–20°N, 140°E–100°W; 4,833 grid cells). The two historical strong El Niño events are shown separately. b,d,f, Relative correlation decrease: 100 × (r_reference − r_2023–2024)/r_reference, using each historical event and their equal-event mean. Positive values indicate deterioration; negative values indicate improvement. All right-hand panels share the same vertical scale. Lead 9 is unavailable and remains blank/NA; no nearest-lead substitution is used.

Forecast initialization equals the verifying month minus the labelled lead; the internal S coordinate must match that initialization and L must equal lead + 0.5. Monthly anomalies are calculated against 1992–2020, with separate lead-dependent model and observed climatologies using identical available calendar months. DJF is averaged before spatial correlation. The same ocean mask is used for all 72 valid event–variable–lead correlations. Eight leads within each of three events are repeated evaluations, not independent events; reference curves are sensitivity comparisons, not confidence intervals. This is a controlled single-model comparison, not a multimodel attribution of MHW forecast-skill loss.